

# Xylophone Capstone

## Instrument-Maker Print Packet

Build packet folder: /mnt/c/Users/Tony/Documents/GitHub/xylophone

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This packet is the printable companion to the build folder. Take it shopping or into the shop. Tear sheets at page breaks.

## File Map

| File                | Purpose                                                                                   |
|---------------------|-------------------------------------------------------------------------------------------|
| design.md           | Project intent, catalog metadata, assumptions, and validation plan.                       |
| bom.csv             | Starter bill of materials with part categories, quantities, drawing refs, and notes.      |
| sourcing.csv        | Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks. |
| cut-list.csv        | Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.     |
| drawing-brief.md    | Manufacturing drawing and technical product sketch brief.                                 |
| assembly-manual.md  | Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.  |
| validation.csv      | Target/measured values, tolerance, environment, result, and tuning/build action log.      |
| supplier-rfq.md     | Supplier email/request-for-quote starter.                                                 |
| visual-bom-brief.md | Art direction for an image-forward visual BOM.                                            |
| wolfram-starter.wl  | Wolfram starter for physics, optimization, visualization, and validation.                 |
| README.md           | Project artifact.                                                                         |
| family-spec.csv     | Project artifact.                                                                         |
| jig-decision.md     | Project artifact.                                                                         |
| photo-shotlist.md   | Project artifact.                                                                         |
| resources.md        | Project artifact.                                                                         |
| risks.md            | Project artifact.                                                                         |

# design.md

*Project intent, catalog metadata, assumptions, and validation plan.*

## Xylophone — Design

A 25-bar chromatic xylophone, C5 to C7, in Padauk on a parametric trestle frame. Free-free flat-bar geometry, 12th-overtone tuning convention, optional quarter-wave closed-pipe resonators (deferred). Designed deliberately as the \*simpler\* sibling of a future marimba sister repo: every place this design chooses the simpler path, the marimba-equivalent is documented inline so the marimba scaffold inherits cleanly.

### Intent

Ship a parametric, validator-clean, public-readable Mode A build packet for a serious wooden tuned-bar idiophone — small enough to build in one shop session, honest enough about the physics that the predicted frequencies and the measured frequencies will disagree on first cut and the workflow can correct them, structured enough that a marimba follow-on can reuse the family table, the validation rows, the suspension geometry, the SolidWorks master-layout convention, and the empirical-correction guard rules.

The reader should leave with: (a) a parametric design table whose inputs are wood species, bar thickness, target pitch, and frame geometry; (b) a defensible first-order frequency prediction per note; (c) a measured-tuning workflow that closes the loop on the prediction; (d) a clear "this is what turns into marimba next" map.

### Governing Model

Free-free Euler-Bernoulli transverse beam vibration, first mode:

$$f_1 = 1.028 \times (h / L^2) \times \sqrt{E / \rho}$$

Symbols and units (SI):

- $h$  — bar thickness (m)
- $L$  — bar length (m)
- $E$  — Young's modulus along grain (Pa)

- $\rho$  — density (kg/m<sup>3</sup>)
- $f_1$  — first-mode frequency (Hz)

Reference: ``acoustic-models.md`` Free-Free Beams section. The formula follows from  $K_{\text{free\_free}} \approx 6.36 \times K_{\text{cantilever}}$  applied to the cantilever form  $f_1 \approx 0.162 \times (h/L^2) \times \sqrt{(E/\rho)}$ . Suspension nodes for the first free-free mode are at **22.4% and 77.6% of L** — every cord-hole in ``family-spec.csv`` is placed there.

**First-order estimate caveat.** Per the ``acoustic-models.md`` empirical-correction guard rules, K2 corrections, scale-offset tables, and end-corrections **do not apply** to bar idiophones. Material K-constants from the cantilever wood table are species-specific and only valid for the free-free / fixed-free regime — they propagate to this packet via the 6.36× factor above. Every predicted frequency in ``family-spec.csv`` is flagged as *first-order*; ``validation.csv`` is the mechanism that closes the gap with measured data after first prototype.

**Xylophone tuning convention.** Bars are tuned to align the *12th overtone* (3:1 frequency ratio above the fundamental — actually the second flexural mode tuned to a perfect 12th of the fundamental in the concert convention; first-order math reports  $f_1$  only). Marimba tunes the second mode to a 4:1 ratio (two octaves) by parabolic arch undercut. **This packet uses no arch undercut** — bars are flat-bottom rectangular blanks, end-shaved for fundamental tuning. The 12th-overtone target is documented as a *measurement target* in ``validation.csv``, not a CAD geometry — adjust at the bench.

- > Marimba-equivalent: replace the flat-bottom geometry with a parabolic
- > CNC arch on the underside (``acoustic-models.md``:
- >  $\text{center\_thickness} = \text{edge\_thickness} \times (\text{target\_freq} / \text{flat\_bar\_freq})^2$ ,
- > minimum center 0.25"). Replace the 12th-overtone target with the
- > 4:1 octave-and-a-fifth target. Replace the optional resonators with
- > mandatory tuned tubes.

## Family Plan

Two octaves chromatic, C5 to C7, 25 bars. Family geometry held

constant: bar width `W = 1.500"`, bar thickness `h = 0.875" (7/8" flat stock). Only `L` varies — solved per note from the governing model. See `family-spec.csv` for the full table.

Predicted lengths run from **16.683"** at C5**\*\*** to **8.342"** at C7**\*\***.

These are noticeably longer than a concert xylophone of the same range because the design has no arch undercut — the arch is what lets a concert xylo hit the same pitch with a 10–12" bar. The flat-bar tradeoff is honest, parametric, and easier to first-cut.

Width and thickness are constants in this v1 family. Production xylophones often taper width across the range; that's a deferred optimization, called out in Open Assumptions.

- > Marimba-equivalent: keep width constant within an octave but step
- > width up at register boundaries (low marimba bars are wider). Add an
- > arch-depth column. Add a resonator-tube-bore column.

## Wood Species

Three-tier shortlist; **Padauk** is the spec's primary**\*\*** because it is sustainably sourced, has predictable acoustic properties, machines cleanly, and is widely available in 4/4 quartersawn:

| Tier               | Species           | E (GPa) | ρ (kg/m³) | √(E/ρ) (m/s) | Notes                                              |
|--------------------|-------------------|---------|-----------|--------------|----------------------------------------------------|
| Primary            | African Padauk    | 12.6    | 745       | 4112         | Spec default. Stable, predictable, sustainable.    |
| Traditional        | Honduras Rosewood | 14.1    | 1000      | 3756         | Concert standard. <b>CITES II</b> — sourcing risk. |
| Educational/budget | Hard Maple        | 12.6    | 705       | 4229         | Cheapest, most available, dampest sound.           |

Switching the primary requires a single edit in `xylophone-design-table.xlsx` (input cell `wood\_species`); the family table re-solves automatically. The validator runs on the table's current state, not on a hardcoded snapshot.

- > Marimba-equivalent: same shortlist, but most concert marimbas use
- > Padauk or Honduras Rosewood; Birch is the educational tier.

## Tuning Workflow

The validation loop has eight rows in `validation.csv`. Two are prediction-only; six require a tuner and a chromatic environment log:

1. Rough-cut prediction (paper, before any cut).
2. Post-cut fundamental measurement (immediately after bandsaw + plane).
3. Post-shave measurement (after end-shaving for fundamental tune).
4. Post-finish measurement (after oil/wax — finish adds mass, drops pitch).
5. Post-resonator-install measurement (when resonators are installed; deferred).
6. 30-day moisture re-measure (drift after first shop cycle).
7. 60-day moisture re-measure (drift after a full season cycle).
8. Full-octave evenness check (every C across both octaves on the same day).

Each row carries `tuner\_make\_model`, `temp\_F`, `humidity\_RH\_percent`, and `cents\_target` columns matching `record\_measurement.py`'s schema. A measured Hz feeds `record\_measurement.py --packet . --note-id <X> --measured-hz <f>` and the per-family corrections database absorbs the delta.

## Resonator Decision

**\*\*Optional, deferred to v2 build.\*\*** The L-per-note math is computed in `family-spec.csv` (column `resonator\_l\_in\_optional`) using the quarter-wave closed-pipe formula with end correction:

$$L_{\text{resonator}} = c / (4 f) - 0.82 \times \text{bore}$$

with `c = 343 m/s` and `bore = 1.500"`. Lengths run from **\*\*5.22"** at C5\*\* down to **\*\*0.38"** at C7\*\*. The high-end values are too short to build practically — the production decision will be: tune resonators on bars at C5 through approximately G5 (lengths > 3"), and skip resonators on the upper octave. This decision is made when resonators are installed, not now.

> Marimba-equivalent: resonators are mandatory; the bore is bigger  
> (typically 2.0–3.0"); each tube is plugged at the bottom and tuned  
> at the top end. Marimba tubes are also typically *\*not\** trimmed by  
> the 0.82×bore end-correction directly — they're trimmed by ear at  
> the bench.

## Frame Layout

4-leg parametric trestle frame, two suspension rails running the full length of the bar fan. Rail length sized to ``max(family-spec L) + 4" margin = 16.683 + 4 = 20.683"` rounded up to `**21**`. Leg height `**30**` (standard standing-play height; sit-down play needs a swap to 24"). Cord suspension with ``1/8"` paracord through the node holes; cord termination at the rail ends with a tied-loop and finish washer.

Frame parts are CNC candidates (rails, leg blanks, gussets) — see ``cnc/operations.csv``. Bars are explicitly out of CNC scope: bandsaw to rough → surface plane to thickness → drum-sand surfaces → drill node holes (jig) → end-shave for tuning (manual). The CNC plan states this exclusion in the setup sheet.

- > Marimba-equivalent: longer frame (4-octave marimba is 5+ feet),
- > resonator wells underneath each bar pair, height-adjustable legs,
- > traveling-cart casters.

## Mallets

Hard rubber → lexan → rosewood-head shortlist. Hard mallets are appropriate for xylophone (the percussive, glassy attack is part of the instrument's voice). They are `**harder than marimba mallets**` and `*can*` crack a softer-wood bar — see ``risks.md`` entry ``bar-cracking-from-hard-mallets``. Standard practice: yarn-wrapped mallets for low octave, lexan/rubber for upper. v1 packet ships with hard rubber as the recommended starter; lexan and rosewood are swap-ins.

- > Marimba-equivalent: yarn-wrapped soft mallets across the entire
- > range. The bars are softer (Honduras Rosewood is a hardwood but
- > with arch undercut the center is thin); soft mallets prevent
- > cracking and produce a warmer overtone profile.

## Marimba-Equivalent Hooks (summary map)

| Decision | Xylophone (this packet) | Marimba (sister repo, deferred) |  
 |-----|-----|-----|  
 | Bar geometry | Flat-bottom rectangular | Parabolic arch undercut on underside |  
 | Tuning ratio target | 12th overtone (3:1) | 4:1 (two octaves) |  
 | Resonators | Optional, deferred | Mandatory, tuned per bar |  
 | Bar width policy | Constant 1.500" across range | Step up at register boundaries |  
 | Mallet hardness | Hard rubber → lexan → rosewood | Yarn-wrapped soft → medium |  
 | Suspension cord | 1/8" paracord | 3/16" or 1/4" (heavier bars) |  
 | Frame length | 21" rail | 60"+ rail, multi-octave |  
 | Wood primary | Padauk | Padauk or Honduras Rosewood |

A future `marimba/` repo can `cp -r xylophone/\* marimba/` and edit exactly these eight rows in the design table to bootstrap.

## Open Assumptions

1. **No measured data yet.** Every frequency in `family-spec.csv` is a first-order prediction. First measurement after first prototype is the single most important next step. (`validation.csv` row 2.)
2. **Wood species.** Padauk is the primary by spec choice; Honduras Rosewood and Hard Maple swap in by changing one cell in the workbook. CITES sourcing for HR is not analyzed in this packet.
3. **Family range.** C5–C7 chromatic is the chosen default. F4–F6 diatonic and G4–G6 diatonic remain documented alternatives — they trade range for shorter bar stock and a smaller frame.
4. **Bar thickness held constant at 7/8".** A real production xylo often steps thickness up in the low octave to keep bars from being absurdly long. This packet keeps thickness constant for parametric simplicity and accepts the 16.7" C5 bar.
5. **Resonator strategy.** The L-per-note math is computed but resonators are not specified in the BOM. Adding them is additive, not breaking — the frame design leaves vertical clearance.
6. **CAD.** SolidWorks master-layout part is deferred; the `cad/` folder has a stub README pointing at the tongue-drum convention and the existing `xylophone-design-table.xlsx`.
7. **Mallet supplier.** v1 lists generic hard-rubber mallets; specific supplier (Innovative Percussion vs Encore vs Vic Firth) is not

selected.

These eight rows are exactly the assumptions the handoff report flags. Each becomes a `validation.csv` or `bom.csv` line item once the gating decision is made.



# bom.csv

Starter bill of materials with part categories, quantities, drawing refs, and notes.

| part_id        | part_name                            | qty                            | unit                 | estimated_cost | supplier_candidate    | notes                                     |
|----------------|--------------------------------------|--------------------------------|----------------------|----------------|-----------------------|-------------------------------------------|
| BAR-STOCK      | Padauk 4/4 quartersawn               | 1 surfaced 4 sides, KD 6-8% MC | board-foot-bundle-25 | 180.00         | Bell Forest Products  | sized to yield 25 bars 8.3-16.7in long    |
| FRAME-RAIL     | Hard maple 8/4 rail blank            | 2 24in x 2.5in x 1.5in         | piece                | 28.00          | Bell Forest Products  | two suspension rails; rough oversize      |
| FRAME-LEG      | Hard maple 8/4 leg blank             | 4 32in x 1.75in x 1.75in       | piece                | 18.00          | Bell Forest Products  | four legs; rough oversize                 |
| FRAME-GUSSET   | Baltic birch ply 1/2in 12x12         | 12 panel                       | piece                | 14.00          | Lee Valley            | two corner gussets per end                |
| CORD-NODE      | 1/8 inch paracord black              | 100ft hank                     | hank                 | 9.00           | Lee Valley            | node-suspension cord                      |
| CORD-WASHER    | Brass finish washer 1/8in            | 1 cord 25-pack                 | pack                 | 8.00           | Lee Valley            | cord termination                          |
| SCREW-FRAME    | #10 x 2in zinc wood screw            | 1w 50-pack                     | pack                 | 7.00           | Lee Valley            | frame assembly                            |
| GLUE-FRAME     | Titebond III 16oz                    | 1                              | bottle               | 12.00          | Lee Valley            | frame joinery                             |
| FINISH-OIL     | Tung oil pure 16oz                   | 1                              | bottle               | 18.00          | Lee Valley            | bar and frame finish                      |
| FINISH-WAX     | Beeswax paste 8oz                    | 1                              | tin                  | 11.00          | Lee Valley            | top coat                                  |
| MALLET-HRD     | Hard rubber xylophone mallet         | 2 mallet pair                  | pair                 | 32.00          | Innovative Percussion | primary mallet for upper octave           |
| MALLET-MED     | Yarn-wrapped medium rubber mallet    | 2 mallet pair                  | pair                 | 38.00          | Innovative Percussion | alt mallet for low octave                 |
| TUNER-CHROMA   | Korg OT-120 orchestral tuner         | 1 tuner                        | unit                 | 140.00         | Sweetwater            | validation row 2-8 device of record       |
| HYGRO-SHOP     | Combo hygrometer-thermometer digital | 1                              | unit                 | 22.00          | Lee Valley            | validation env log                        |
| MOIST-METER    | Pin-style moisture meter             | 1                              | unit                 | 38.00          | Lee Valley            | verify 6-8% MC at glue-up                 |
| SANDPAPER-KIT  | Aluminum oxide grit 80-220           | 220 assortment                 | kit                  | 18.00          | Lee Valley            | bar surface prep                          |
| DRILL-BIT-NODE | 1/8in brad-point bit                 | 2                              | piece                | 4.50           | Lee Valley            | node-hole drilling, jig-mounted           |
| RESONATOR-TUBE | Aluminum tube 1.5in OD 23.6in length | 1                              | piece                | TBD            | McMaster-Carr         | deferred to v2 build; sized for resonator |

# sourcing.csv

Supplier/search tracker with specs, price/date fields, lead time, substitutes, and risks.

| part_id        | supplier_candidate    | required_spec                  | search_terms                           | lead_time_days_est | notes                                        |
|----------------|-----------------------|--------------------------------|----------------------------------------|--------------------|----------------------------------------------|
| BAR-STOCK      | Bell Forest Products  | African Padauk 4/4 quarters    | padauk 4/4 quartersawn                 | 8                  | primary; ask for low-shake low-pin selection |
| BAR-STOCK      | West Penn Hardwoods   | African Padauk 4/4 quarters    | padauk 4/4 quartersawn                 | 10                 | backup if Bell is OOS                        |
| BAR-STOCK      | Cook Woods            | African Padauk billet select   | padauk billet quartersawn              | 14                 | third option; pricier but selectable boards  |
| FRAME-RAIL     | Bell Forest Products  | Hard maple 8/4 KD straight     | hard maple 8/4 straight grain          | 7                  | two boards 30in min length                   |
| FRAME-LEG      | Bell Forest Products  | Hard maple 8/4 KD              | hard maple 8/4                         | 7                  | four leg blanks                              |
| FRAME-GUSSET   | Lee Valley            | Baltic birch plywood 1/2in B   | Baltic birch 1/2in plywood 12x6        | 2                  | or substitute Home Depot if local            |
| CORD-NODE      | Lee Valley            | Type II 1/8in paracord 100ft   | paracord 1/8 100ft                     | 5                  | black or natural                             |
| CORD-WASHER    | Lee Valley            | Brass finish washer fits 1/8in | brass finish washer 1/8 cord           | 5                  | or McMaster-Carr 91068A350                   |
| SCREW-FRAME    | Lee Valley            | #10 x 2in steel wood screw     | wood screw #10 2in zinc                | 5                  | 50-pack minimum                              |
| GLUE-FRAME     | Lee Valley            | Titebond III 16oz              | titebond III 16oz                      | 5                  | waterproof PVA, longer open time             |
| FINISH-OIL     | Lee Valley            | Pure tung oil food-safe 16oz   | pure tung oil 16oz                     | 5                  | food-safe so the kid players are fine        |
| FINISH-WAX     | Lee Valley            | Beeswax paste furniture finish | beeswax furniture paste 8oz            | 5                  | top coat over tung                           |
| MALLET-HRD     | Innovative Percussion | Hard rubber xylophone mallet   | hard rubber xylophone mallet           | 7                  | alt: Encore EM-1, Vic Firth M134             |
| MALLET-MED     | Innovative Percussion | Yarn-wrapped medium marimba    | yarn mallet marimba mallet             | 7                  | alt: Encore EX-1                             |
| TUNER-CHROMA   | Sweetwater            | Korg OT-120 orchestral tuner   | Korg OT-120 orchestral tuner           | 3                  | or Peterson StroboPlus HD if available       |
| HYGRO-SHOP     | Lee Valley            | Digital hygrometer thermometer | digital hygrometer thermometer         | 3                  | target accuracy +/-3% RH                     |
| MOIST-METER    | Lee Valley            | Pin-style moisture meter wood  | pin moisture meter wood                | 5                  | or General MM50                              |
| SANDPAPER-KIT  | Lee Valley            | Aluminum oxide sanding sheet   | aluminum oxide sandpaper               | 3                  | hand-sand bar surfaces post-plane            |
| DRILL-BIT-NODE | Lee Valley            | Brad point drill bit 1/8in     | brad point bit 1/8 inch                | 5                  | two for spares                               |
| RESONATOR-TUBE | McMaster-Carr         | 6061 aluminum tube 1.5in OD    | 6061 aluminum tube 1.5in OD 0.063 wall | 5                  | deferred; quote at v2 build time             |

# cut-list.csv

Rough/final stock sizes, material, grain/orientation, operations, yield, and offcuts.

| part_id | part_name             | qty | rough_dimensions   | final_dimensions       | grain_orientation    | notes                                     |
|---------|-----------------------|-----|--------------------|------------------------|----------------------|-------------------------------------------|
| BAR-C5  | Bar C5                | 1   | 18.0 x 1.625 x 1.0 | 16.683 x 1.500 x 0.875 | longitudinal-quarter | longest bar; cut from longest blank       |
| BAR-Cs5 | Bar C#5               | 1   | 17.5 x 1.625 x 1.0 | 16.208 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-D5  | Bar D5                | 1   | 17.0 x 1.625 x 1.0 | 15.747 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Ds5 | Bar D#5               | 1   | 16.5 x 1.625 x 1.0 | 15.299 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-E5  | Bar E5                | 1   | 16.0 x 1.625 x 1.0 | 14.863 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-F5  | Bar F5                | 1   | 15.5 x 1.625 x 1.0 | 14.440 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Fs5 | Bar F#5               | 1   | 15.0 x 1.625 x 1.0 | 14.029 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-G5  | Bar G5                | 1   | 15.0 x 1.625 x 1.0 | 13.630 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Gs5 | Bar G#5               | 1   | 14.5 x 1.625 x 1.0 | 13.242 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-A5  | Bar A5                | 1   | 14.0 x 1.625 x 1.0 | 12.865 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-As5 | Bar A#5               | 1   | 13.5 x 1.625 x 1.0 | 12.498 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-B5  | Bar B5                | 1   | 13.5 x 1.625 x 1.0 | 12.143 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-C6  | Bar C6                | 1   | 13.0 x 1.625 x 1.0 | 11.797 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Cs6 | Bar C#6               | 1   | 12.5 x 1.625 x 1.0 | 11.461 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-D6  | Bar D6                | 1   | 12.5 x 1.625 x 1.0 | 11.135 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Ds6 | Bar D#6               | 1   | 12.0 x 1.625 x 1.0 | 10.818 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-E6  | Bar E6                | 1   | 11.5 x 1.625 x 1.0 | 10.510 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-F6  | Bar F6                | 1   | 11.5 x 1.625 x 1.0 | 10.211 x 1.500 x 0.875 | longitudinal-quarter |                                           |
| BAR-Fs6 | Bar F#6               | 1   | 11.0 x 1.625 x 1.0 | 9.920 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-G6  | Bar G6                | 1   | 10.5 x 1.625 x 1.0 | 9.638 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-Gs6 | Bar G#6               | 1   | 10.5 x 1.625 x 1.0 | 9.363 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-A6  | Bar A6                | 1   | 10.0 x 1.625 x 1.0 | 9.097 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-As6 | Bar A#6               | 1   | 9.5 x 1.625 x 1.0  | 8.838 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-B6  | Bar B6                | 1   | 9.5 x 1.625 x 1.0  | 8.586 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| BAR-C7  | Bar C7                | 1   | 9.0 x 1.625 x 1.0  | 8.342 x 1.500 x 0.875  | longitudinal-quarter |                                           |
| RAIL-L  | Suspension rail left  | 1   | 22.0 x 2.5 x 1.5   | 21.000 x 2.250 x 1.250 | longitudinal         | both rails identical                      |
| RAIL-R  | Suspension rail right | 1   | 22.0 x 2.5 x 1.5   | 21.000 x 2.250 x 1.250 | longitudinal         |                                           |
| LEG-FL  | Front-left leg        | 1   | 32.0 x 1.75 x 1.75 | 30.000 x 1.500 x 1.500 | longitudinal         | chamfer top edge for cord clearance       |
| LEG-FR  | Front-right leg       | 1   | 32.0 x 1.75 x 1.75 | 30.000 x 1.500 x 1.500 | longitudinal         |                                           |
| LEG-RL  | Rear-left leg         | 1   | 32.0 x 1.75 x 1.75 | 30.000 x 1.500 x 1.500 | longitudinal         |                                           |
| LEG-RR  | Rear-right leg        | 1   | 32.0 x 1.75 x 1.75 | 30.000 x 1.500 x 1.500 | longitudinal         |                                           |
| GUSSET  | Corner gusset         | 4   | 7.0 x 7.0 x 0.5    | 6.000 x 6.000 x 0.500  | plywood-9-ply        | 45-degree triangle gussets at leg-to-rail |

# drawing-brief.md

*Manufacturing drawing and technical product sketch brief.*

## Xylophone — Drawing Brief

What the SVG and CAD drawings need to show, in priority order. The drawings live in `drawings/` (per-family-member SVGs and a frame elevation/plan); the CAD master lives in `cad/` once built.

### DR-01: Bar template (parametric)

A single dimensioned drawing template that varies per family member.

**\*\*Critical dimensions:\*\***

- Bar length `L` (from `family-spec.csv` per note).
- Bar width `W = 1.500`".
- Bar thickness `h = 0.875`" (section view).
- Node-hole positions: low-side at  $0.224 \times L$ , high-side at  $0.776 \times L$ . Hole diameter  $1/8$ ".
- Strike line at center ( $L/2$ ) — annotation only, not a feature.
- Edge round-over  $1/16$ " on all four long edges.

Title block must show: drawing ID, note name, family `XYL`, scale (1:2 reads cleanly for the long bars), revision, date, designer, material, finish.

The drawing template's only variable is `L` and the two derived hole positions. A generator should be able to produce 25 SVGs from the family table by string substitution alone.

### DR-02: Frame elevation

Side view of the assembled frame (no bars). Shows:

- Two suspension rails,  $21.000" \times 2.250" \times 1.250$ ", mounted on top of the legs.
- Four legs,  $30.000" \times 1.500" \times 1.500$ ". Chamfered top edge.
- Corner gussets,  $6.000 \times 6.000 \times 0.500$ " plywood, on the inside face of each leg-rail joint.

- Cord-suspension geometry — schematic, not dimensioned.
- Overall envelope: 21" wide, 30" tall.

## DR-03: Frame plan

Top view of the frame with bars laid in suspension. Shows:

- Both rails parallel, 6.5" inside-to-inside.
- 25 bar positions in the family-table order, lowest at the left.
- Inter-bar gap: 0.250" minimum.
- Total bar-array length: ~17" (driven by the longest bar plus gaps).
- Cord routing: alternating side-to-side through node holes.

## DR-04: Mallet detail

Three-view of the recommended hard-rubber mallet (from `bom.csv`). Shaft length, head diameter, head shape. **\*\*Reference only\*\*** — the mallets are purchased, not built. Goes on the visual BOM, not in the build instructions.

## Style notes

- All dimensions in inches. Decimal, three places (`16.683`), consistent with `family-spec.csv` and `cut-list.csv`.
- Datum chain: bar drawings reference left bar end; frame drawings reference left-rear leg base.
- Tolerances: bars  $\pm 1/64$ ", frame  $\pm 1/32$ " (frame loose on purpose — bars do the precision work).
- File names: `drawings/DR-01-bar-<note>.svg` per bar (or a single `drawings/template.svg` until the per-bar instances are generated). Frame drawings: `drawings/DR-02-frame-elevation.svg`, `drawings/DR-03-frame-plan.svg`. Mallet: `drawings/DR-04-mallet.svg`.

## Generator hookup

A future invocation of `scripts/generate\_drawings.py` against this packet will populate `drawings/DR-01-bar-\*.svg` from `family-spec.csv`.

Until that runs, `drawings/template.svg` is the placeholder showing the title-block layout and the datum chain so the generator's output matches in style.

# assembly-manual.md

*Shop-facing sequence, tools, fixtures, safety, tuning, finishing, and maintenance notes.*

## Xylophone — Assembly Manual

14 shop steps, in order. Checkboxes are blank for shop use; tick them off as you go. Estimated total bench time: **\*\*3 working days\*\*** with overnight glue cures.

### Materials and tools required

Read `bom.csv` and `sourcing.csv` end-to-end before starting. The critical-path items: Padauk bar stock at 6–8% MC, Korg OT-120 tuner, 1/8" brad-point bit, Titebond III, tung oil, and the two jigs from `jig-decision.md` (node-hole drilling jig, end-shaving sled).

### Step 1 — Verify moisture content

- ☐ Pin-meter the Padauk bar stock at three points along each board. Target 6.0–8.0% MC.
- ☐ Log temp + RH in the shop in `validation.csv` row V1.
- ☐ If MC > 8%: stack with stickers in the shop for two weeks and remeasure. **\*\*Do not proceed until MC is 6–8%.\*\***

### Step 2 — Mill blanks to rough cut-list dimensions

- ☐ Cross-cut each bar blank to its `rough\_dimensions\_in[0]` length (per `cut-list.csv`). Add 1" to the longest blanks for safety.
- ☐ Rip to 1.625" rough width.
- ☐ Joint one face and one edge.
- ☐ Plane to 1.0" rough thickness.

### Step 3 — Final mill bars

- ☐ Plane to **\*\*0.875"\*\*** final thickness.
- ☐ Rip to **\*\*1.500"\*\*** final width.
- ☐ Cross-cut to the `final\_dimensions\_in[0]` length per

`cut-list.csv`. **\*\*Cut long by 1/16"** — final length is set in the tuning step.

## Step 4 — Surface prep

- [ ] Drum-sand or hand-sand each bar 80 → 120 → 180 → 220.
- [ ] Ease all four long edges with a 1/16" round-over.
- [ ] Mark each bar's note name on the underside in pencil.

## Step 5 — Drill node holes (jig)

- [ ] Set up the node-hole drilling jig from `jig-decision.md`.
- [ ] For each bar: set both flip-stops to the bar's `hole\_pos\_low\_in` and `hole\_pos\_high\_in` from `family-spec.csv`.
- [ ] Drill the low-side hole, flip stops, drill the high-side hole.
- [ ] Chamfer both holes ~0.5mm with a countersink to prevent cord chafing.

## Step 6 — First measurement

- [ ] Suspend each bar from a piece of test-cord through both node holes.
  - [ ] Strike with the hard rubber mallet, measure with the Korg OT-120.
  - [ ] Log result for each bar in `validation.csv` row V2.
  - [ ] **\*\*Expected outcome:\*\*** bars read 5–25 cents *\*low\** of target.
- This is normal — final length is shaved in the next step.

## Step 7 — Tune by end-shaving

- [ ] On the end-shaving sled: shave equal amounts from each bar end in 1/64" increments. Symmetric shaving keeps the node positions centered.
- [ ] After each pass, re-suspend, re-strike, re-measure. Stop when cents-actual is within  $\pm 2$  cents of target.
- [ ] Log result in `validation.csv` row V3.
- [ ] If a bar overshoots (reads sharp): **\*\*do not add material\*\***.



Either accept the +cents or scrap the bar.

## Step 8 — Mill frame parts

- [ ] Mill the two suspension rails to `21.000 × 2.250 × 1.250``.
- [ ] Mill four legs to `30.000 × 1.500 × 1.500``. Chamfer the top edge of each leg to clear the suspension cord.
- [ ] Cut four corner gussets from 1/2" Baltic birch plywood at `6.000 × 6.000``, hypotenuse to hypotenuse (i.e., 45-degree triangles).

## Step 9 — Frame dry-fit

- [ ] Dry-assemble the frame: legs into rails via the gussets.
- [ ] Verify rail-to-rail spacing is 6.5" inside-to-inside (matches the bar width plus suspension margin).
- [ ] Verify legs are square to rails.

## Step 10 — Frame glue-up

- [ ] Apply Titebond III to all gusset-to-frame joints.
- [ ] Drive #10 × 2" wood screws through pilot holes (Padauk and hard maple both want pilots).
- [ ] Clamp 24 hours. Do not flex the frame during cure.

## Step 11 — Apply finish

- [ ] Wipe each bar with mineral spirits to remove sanding dust.
- [ ] Apply pure tung oil with a lint-free rag, three coats. Each coat dries 24 hours; sand lightly between coats with 320.
- [ ] Apply finish to the frame the same way (two coats is enough).
- [ ] **\*\*24 hours after the last bar coat:\*\*** re-measure each bar. Log in `validation.csv` row V4. Expected drift: -2 to -4 cents from finish mass loading. If drift exceeds -5 cents, end-shave again to compensate.

## Step 12 — Top-coat and cure

- ☐ Apply beeswax paste over the cured tung oil; buff out.
- ☐ Allow seven days at shop ambient before play-testing.

## Step 13 — Suspend bars

- ☐ Cut paracord to length: rail length  $\times$  2 + 12" tail per rail.
- ☐ Tie a stopper knot at each rail end.
- ☐ Thread cord through node holes in C5–C7 order, alternating bar-to-cord-to-bar.
- ☐ Terminate at the second rail with brass finish washers and a tied loop.

## Step 14 — Final validation

- ☐ Photograph the finished assembly per `photo-shotlist.md` shots 1, 2, 4, 5, 6, 7, 10.
- ☐ Run validation row V8 (full-octave evenness): measure C5, C6, C7 in one session. Document spread.
- ☐ Schedule validation rows V6 (30-day) and V7 (60-day) on the shop calendar.
- ☐ Mark this packet as `built` in the master catalog.

validation.csv

Target/measured values, tolerance, environment, result, and tuning/build action log.

| row_id | note_id  | phase                  | target_hz                | measured_hz | cents_target | cents_actual     | tolerance | maker_model    | humidity | temp | environment                          | notes                     | result                      |
|--------|----------|------------------------|--------------------------|-------------|--------------|------------------|-----------|----------------|----------|------|--------------------------------------|---------------------------|-----------------------------|
| V1     | ALL      | prediction-paper       | per family-spec          |             | 0            |                  |           | N/A            | 68       | 45   | design table                         | pre-cut                   | first-order Euler-Bernoulli |
| V2     | A5       | post-cut               | 880.00                   |             | 0            |                  |           | Korg OT-12068  | 45       |      | shop bench                           | immediately after sanding | acceptance; expected        |
| V3     | A5       | post-shave             | 880.00                   |             | 0            |                  |           | Korg OT-12068  | 45       |      | shop bench                           | after end-shave           | may be 64 cents; expected   |
| V4     | A5       | post-finish            | 880.00                   |             | -3 to 0      |                  |           | Korg OT-12068  | 45       |      | shop bench                           | 24 hours after finish     | add class; expected         |
| V5     | A5       | post-resonator         | 880.00                   |             | 0            |                  |           | Korg OT-12068  | 45       |      | shop, with resonator tube            | installed at 0.2          | tail wave level             |
| V6     | ALL      | 30-day-moisture        | per family-spec          |             | within +/-5  |                  |           | Korg OT-120TBD | TBD      |      | 30 days post-finish; log temperature | required                  | any bar drifting beyond     |
| V7     | ALL      | 60-day-moisture        | per family-spec          |             | within +/-5  |                  |           | Korg OT-120TBD | TBD      |      | 60 days post-finish; log temperature | second R                  | disturbance check; if       |
| V8     | C5-C6-C7 | full-octave-evaluation | 520.25, 1046.50, 2093.00 |             | within +/-5  | across the three |           | Korg OT-12068  | 45       |      | all three Cs measured in one session | for 10 cents              | spread                      |

# supplier-rfq.md

Supplier email/request-for-quote starter.

## Xylophone — Supplier RFQ

Template — paste into supplier email or web form. The bar-stock RFQ is the only critical-path RFQ; everything else is COTS from `sourcing.csv` and does not need a custom quote.

---

**Subject:** Padauk bar-stock quote — 25-bar xylophone build

Hello,

I'm building a 25-bar chromatic xylophone (C5–C7) and I'd like a quote on Padauk lumber for the bar set. Specs:

- **Species:** African Padauk (*Pterocarpus soyauxii*) — *not* Burma Padauk.
- **Form:** 4/4 lumber (15/16" rough, surfaced 4 sides to roughly 3/4"–7/8" final), or fall-back to S2S if S4S is unavailable.
- **Grain:** **Quartersawn** strongly preferred (radial grain on the bar face). Riftsawn acceptable on 25–35% of the order. Flatsawn rejected — bars will warp.
- **Moisture content:** **Kiln-dried to 6–8% MC.** Please measure with a pin meter at multiple points and confirm in writing.
- **Defect tolerance:** No knots, no shake, no end-checks longer than 1/2", no insect damage, no sap pockets. Color variation acceptable.
- **Total board feet:** Approximately **15 BF** to yield 25 bars (longest 17", shortest 9", at 1.5"×7/8") plus 20% waste.
- **Lengths preferred:** Boards in the 36–48" range. Anything shorter than 24" is hard to use.

Please quote:

1. Per-board-foot price for the spec above.
2. Total for 15 BF, FOB your shipping point.
3. Lead time from order to ship.

4. Photographs of two or three representative boards before payment so I can confirm grain orientation and color.

If S4S quartersawn is not available at this volume, please quote for S2S quartersawn at 5/4 thickness instead and I'll mill to 7/8" myself.

Thanks,  
Tony Koop  
tonykoop@gmail.com

---

## RFQ tracking

| Field            | Value                                 |
|------------------|---------------------------------------|
| RFQ ID           | XYL-RFQ-001                           |
| Spec sheet       | `bom.csv` row `BAR-STOCK`             |
| Sourcing options | `sourcing.csv` rows for `BAR-STOCK`   |
| Sent date        | TBD                                   |
| Quote received   | TBD                                   |
| Decision         | TBD                                   |
| Order placed     | TBD                                   |
| Received in shop | TBD                                   |
| MC at receipt    | TBD (target 6–8%; measure on arrival) |

## Decision criteria

Pick the supplier whose quote yields **best** (price × delivery × grade match). Bell Forest Products is the default expectation; West Penn and Cook are the named backups in `sourcing.csv`.

## What to do if no supplier can deliver to spec

Fall-back order: switch the spec's primary species in `xylophone-design-table.xlsx` from Padauk to Hard Maple (Tier 3 in `design.md`). Re-run the family table — predicted lengths shift slightly because Maple's  $\sqrt{E/p}$  is 4229 m/s vs Padauk's 4112. Bars end up about 1.4% longer. Update `family-spec.csv` and `cut-list.csv`

accordingly.

# visual-bom-brief.md

Art direction for an image-forward visual BOM.

## Xylophone — Visual BOM Brief

Brief for the visual BOM (the recruiter-facing assembly plate). One page, every line in `bom.csv` represented by a thumbnail and a two-line label.

### Page layout

- **Header.** Project name + version + quote date + total cost.
- **Hero strip** at top: `images/01-hero.jpg` (or the placeholder SVG until captured), full-width.
- **Three columns** of part rows below the hero, top-to-bottom: bar stock + finishing on the left, frame stock + hardware in the middle, mallets + tuning + measurement gear on the right.
- **Footer.** Total cost, link to the GitHub repo, license note.

### Part-row anatomy

Each row:

[thumbnail 60×60] PART\_ID — Part Name

Qty • Unit • Estimated cost • Supplier

One-line spec summary

Thumbnails are pulled from `images/` by part-id slug. When an image is missing, fall back to a category placeholder (a square with the part-id text centered, no decoration).

### Mapping bom.csv → visual BOM

| bom.csv part\_id | Image slug | Notes for plate |

|-----|-----|-----|

| BAR-STOCK | `images/02-bar-fan.jpg` | top-left, biggest cell |

| FRAME-RAIL | `images/04-frame-elevation.jpg` | use frame photo cropped to rails |

| FRAME-LEG | `images/04-frame-elevation.jpg` | crop to legs |

| FRAME-GUSSET | placeholder square | small detail, use category placeholder |  
| CORD-NODE | `images/06-suspension-cord.jpg` | cord visible against bars |  
| CORD-WASHER | placeholder square | |  
| SCREW-FRAME | placeholder square | |  
| GLUE-FRAME | placeholder square | |  
| FINISH-OIL | placeholder square | |  
| FINISH-WAX | placeholder square | |  
| Mallet-HRD | `images/05-mallet-trio.jpg` | crop to hard-rubber mallet |  
| Mallet-MED | `images/05-mallet-trio.jpg` | crop to medium yarn mallet |  
| TUNER-CHROMA | `images/07-tuner-on-bar.jpg` | OT-120 mid-measurement |  
| HYGRO-SHOP | placeholder square | |  
| MOIST-METER | `images/09-moisture-meter.jpg` | meter on Padauk end-grain |  
| SANDPAPER-KIT | placeholder square | |  
| DRILL-BIT-NODE | placeholder square | |  
| RESONATOR-TUBE | placeholder square — deferred | grey out the row; v2 build |

## Style

- Black text on white background. No decorative shading.
- Font: Inter or system-ui. Body 9pt, headings 11pt, totals 13pt.
- Print target: US Letter portrait, 0.5" margins. The visual BOM is the one page that must be printable in a shop.

## Output

The visual BOM is rendered as part of the print packet (`print-packet.html`` and `print-packet.pdf``). The HTML is the source of truth; the PDF is generated from it. When optional PDF deps are missing, the HTML alone is shippable.



# wolfram-starter.wl

*Wolfram starter for physics, optimization, visualization, and validation.*

(\* ::Package:: \*)

(\* Xylophone — instrument-model.wl \*)

(\* Free-free Euler-Bernoulli flat-bar idiophone. First-mode prediction, \*)

(\* node positions, optional quarter-wave closed-pipe resonator, validation \*)

(\* plot scaffolding, audio preview. \*)

(\* \*)

(\* Read references/wolfram-workflow.md before extending this file. \*)

(\* ::Section:: \*)

(\* Parameters \*)

(\* Wood species shortlist — Padauk is the spec primary. Override by editing. \*)

woodPadauk = <| "name" -> "African Padauk", "EGPa" -> 12.6, "rhoKg" -> 745. |>;

woodHondurasRosewood = <| "name" -> "Honduras Rosewood", "EGPa" -> 14.1, "rhoKg" -> 1000. |>;

woodHardMaple = <| "name" -> "Hard Maple", "EGPa" -> 12.6, "rhoKg" -> 705. |>;

(\* Active wood — the only switch that matters at the top of the file. \*)

woodActive = woodPadauk;

(\* Bar geometry constants (held constant within a family in v1). \*)

barWidthIn = 1.5;

barThickIn = 0.875;

inToM = 0.0254;

(\* Resonator geometry. \*)

resonatorBoreIn = 1.5;

soundSpeedAir = 343.0; (\* m/s at ~20 C \*)

endCorrectionFactor = 0.82; (\* per acoustic-models.md, quarter-wave closed pipe \*)

(\* ::Section:: \*)

(\* Governing model \*)

(\* Free-free Euler-Bernoulli first mode:

$f_1 = 1.028 * (h / L^2) * \text{Sqrt}[E/\rho]$

Reference: acoustic-models.md, Free-Free Beams section.

First-order only — K2 and end-corrections do not apply to bar idiophones. \*)

```

soundSpeedSolid[E_, rho_] := Sqrt[(E*10.^9) / rho];

f1FromGeometry[L_, h_, E_, rho_] :=
1.028 * (h / L^2) * soundSpeedSolid[E, rho];

lengthFromTarget[fTarget_, h_, E_, rho_] :=
Sqrt[1.028 * h * soundSpeedSolid[E, rho] / fTarget];

resonatorLength[fTarget_, boreIn_] := Module[{boreM, lM},
boreM = boreIn * inToM;
lM = soundSpeedAir / (4 * fTarget) - endCorrectionFactor * boreM;
lM / inToM
];

(* ::Section:: *)
(* Family table generation *)

chromaticC5toC7Notes = {
{"C5",523.25},{ "C#5",554.37},{ "D5",587.33},{ "D#5",622.25},
{"E5",659.26},{ "F5",698.46},{ "F#5",739.99},{ "G5",783.99},
{"G#5",830.61},{ "A5",880.00},{ "A#5",932.33},{ "B5",987.77},
{"C6",1046.50},{ "C#6",1108.73},{ "D6",1174.66},{ "D#6",1244.51},
{"E6",1318.51},{ "F6",1396.91},{ "F#6",1479.98},{ "G6",1567.98},
{"G#6",1661.22},{ "A6",1760.00},{ "A#6",1864.66},{ "B6",1975.53},
{"C7",2093.00}
};

familyRow[note_String, fHz_?NumericQ] := Module[{Lm, Lin, holeLo, holeHi, Lres},
Lm = lengthFromTarget[fHz, barThickIn * inToM, woodActive["EGPa"], woodActive["rhoKg"]];
Lin = Lm / inToM;
holeLo = 0.224 * Lin;
holeHi = 0.776 * Lin;
Lres = resonatorLength[fHz, resonatorBoreIn];
<| "note" -> note, "fHz" -> fHz, "Lin" -> Lin,
"holeLoIn" -> holeLo, "holeHiIn" -> holeHi,
"resonatorLin" -> Lres |>
];

familyTable := familyRow @@@ chromaticC5toC7Notes;

```

(\* ::Section:: \*)

(\* Manipulate — interactive bar-length explorer \*)

(\* The Manipulate lets you sweep wood/L/h and see f1 in real time.

Useful for sanity-checking first-order predictions and for showing  
players the why-bigger-is-lower-pitched relationship. \*)

```
manipulateBar := Manipulate[
  With[{
    fHz = f1FromGeometry[
      LIn * inToM, hIn * inToM, EChoice, rhoChoice]],
    Column[{
      Style[
        StringForm["f1 = `` Hz", NumberForm[fHz, {6, 2}]],
        Bold, 18],
      Plot[
        f1FromGeometry[L * inToM, hIn * inToM, EChoice, rhoChoice],
        {L, 4, 24}, AxesLabel -> {"L (in)", "f1 (Hz)"},
        PlotRange -> {0, 3000},
        GridLines -> {{LIn}, {fHz}}
      ]
    ]],
    {{LIn, 12.0, "bar length L (in)"}, 4.0, 24.0, 0.05},
    {{hIn, 0.875, "bar thickness h (in)"}, 0.5, 1.5, 0.01},
    {{EChoice, 12.6, "Young's modulus E (GPa)"}, 8.0, 16.0, 0.1},
    {{rhoChoice, 745.0, "density rho (kg/m^3)"}, 600.0, 1100.0, 5.0},
    ControlPlacement -> Top
  ];
```

(\* ::Section:: \*)

(\* Audio preview \*)

(\* Synthesizes a 0.6-second pluck of the predicted fundamental for each  
note in the chromatic table. First-order only — not the actual modal  
spectrum of a real bar, just a confirmation of the predicted f1. \*)

```
audioForNote[fHz_?NumericQ] :=
  AudioPad[
```

```

Sound[SoundNote[None, 0.0]] /. _ ->
AudioGenerator["Sin", 0.6, "Frequency" -> fHz, SampleRate -> 44100],
{0, 0.05}
];

playFamily := AudioJoin @@ (audioForNote#[[2]] & /@ chromaticC5toC7Notes);

(* ::Section:: *)
(* Validation plot scaffold *)

(* Compare predicted f1 (column 1) to measured Hz (column 2) once the
shop log fills in. Until measurements exist, the plot just shows
the predicted line and a residual band. Wire to validation.csv via
record_measurement.py. *)

predictedVsMeasuredPlot[predicted_List, measured_List] :=
ListPlot[
{predicted, measured},
PlotStyle -> {{Thick, Blue}, {PointSize[0.012], Red}},
PlotLegends -> {"predicted", "measured"},
AxesLabel -> {"note index (low → high)", "f1 (Hz)"},
PlotRange -> All,
GridLines -> Automatic
];

centsErrorPlot[predicted_List, measured_List] :=
ListPlot[
1200. * Log[2., measured / predicted],
Filling -> Axis,
PlotRange -> {-30, 30},
AxesLabel -> {"note index", "cents error (measured – predicted)"},
GridLines -> {None, {-5, 0, 5}}
];

(* ::Section:: *)
(* Document entry point *)

(* CreateDocument call lives in the notebook layer, not here. Open this
.wl in Wolfram Desktop, evaluate manipulateBar, then evaluate

```

playFamily to hear the predicted family. \*)

(\* End of file \*)

# README.md

*Project artifact.*

## Xylophone

![Xylophone hero — 25-bar chromatic in Padauk on a parametric trestle frame (placeholder)](images/xylophone-hero-placeholder.svg)

A 25-bar chromatic xylophone, **C5 to C7**, in African Padauk on a parametric trestle frame. Free-free flat-bar geometry, 12th-overtone tuning convention, optional quarter-wave closed-pipe resonators (deferred to v2). This repo is a v4.3 root-mode build packet from the [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) catalogue, designed deliberately as the *simpler sibling* of a future `tonykoop/marimba` repo.

## Background

The xylophone is the higher-pitched cousin of the marimba: same physics (free-free Euler-Bernoulli transverse beam vibration), harder mallets, shorter bars, and a different second-mode tuning convention. This packet ships the bar physics, the wood-species shortlist, the suspension geometry, the validation workflow, and the build process end-to-end. It deliberately *defers* the things that make a marimba hard — the parabolic arch undercut, the mandatory tuned resonators, the multi-octave frame — so the xylophone build is feasible in one shop session and so the marimba follow-on can inherit a clean scaffold rather than starting from scratch.

The root-of-truth design table is `xylophone-design-table.xlsx` at the repo root. Every length, hole position, and resonator length in `family-spec.csv` is solved from the parametric inputs there.

## Design overview

- **Family:** 25 bars, C5 (523.25 Hz) through C7 (2093.00 Hz), chromatic.
- **Bar geometry:** flat-bottom rectangular, **1.500" × 0.875"** cross section held constant across the family. Length varies per note;

see [family-spec.csv](family-spec.csv).

- **Wood:** **African Padauk** (sustainable, predictable acoustic properties, available in 4/4 quartersawn). Honduras Rosewood and Hard Maple are documented swap-ins.
- **Suspension:** 1/8" paracord through node holes drilled at 22.4% and 77.6% of bar length — the first free-free mode's nodes.
- **Tuning:** 12th-overtone convention noted as a *measurement target* on the validation sheet, not a CAD geometry. End-shaving is the production tuning method (no arch undercut — that's marimba).
- **Frame:** parametric 4-leg trestle, 21" wide × 30" tall, hard maple rails and legs with Baltic-birch corner gussets.
- **Resonators:** computed but optional. The L-per-note math is in [family-spec.csv] so adding resonators in v2 is additive.

## Family-table preview (first and last)

| Note  | Target Hz | Bar L (in) | Hole low (in) | Hole high (in) | Resonator L (in) | Mass (oz) |
|-------|-----------|------------|---------------|----------------|------------------|-----------|
| ----- | -----     | -----      | -----         | -----          | -----            | -----     |
| C5    | 523.25    | 16.683     | 3.737         | 12.946         | 5.22             | 9.43      |
| A4 *  | 440.00    | n/a        | n/a           | n/a            | n/a              | n/a       |
| A5    | 880.00    | 12.865     | 2.882         | 9.983          | 2.61             | 7.27      |
| C7    | 2093.00   | 8.342      | 1.869         | 6.473          | 0.38             | 4.71      |

\\* A4 = 440 Hz is the standard tuning reference; the family ranges from C5 upward, so A4 itself is not a member.

Full table: [family-spec.csv](family-spec.csv).

## Why this packet exists

This is the v4.3 challenge run for the [instrument-maker-v4](https://github.com/tonykoop/instrument-maker) skill, exercising the **root-mode** packet contract end-to-end on a new wooden-idiophone family. Specifically:

1. Validates the v4.3 root-mode validator (validate\_packet.py --mode root) against a fresh Mode A repo.
2. Demonstrates the bar-idiophone branch of acoustic-models.md's

Free-Free Beams section with first-order honesty about K2 and end-correction guard rules.

3. Builds the design template that the marimba sister repo will inherit — every "marimba-equivalent: ..." callout in [design.md](design.md) is a hook for the marimba build.

## Sister repos and references

- [tonykoop/instrument-maker](https://github.com/tonykoop/instrument-maker) — parent catalogue, the v4 skill source, references and scripts.
- [tonykoop/tongue-drum](https://github.com/tonykoop/tongue-drum) — closest sister repo: another wooden idiophone, free-free / cantilever beam math, SolidWorks master-layout convention, magazine-baseline attribution. The xylophone README borrows its structure.
- tonykoop/marimba` \*(planned)\* — the natural follow-on; this packet's `design.md` calls out exactly what each marimba decision has to swap in (Marimba-Equivalent Hooks` table).

## Repo layout

xylophone/

- README.md ← this file
- design.md ← governing model, family plan, decisions
- family-spec.csv ← parametric per-note table (25 rows)
- bom.csv, sourcing.csv ← parts and supplier candidates
- cut-list.csv ← shop cut list (bars + frame)
- validation.csv ← measured-tuning workflow (8 rows)
- assembly-manual.md ← 14-step shop manual
- supplier-rfq.md ← bar-stock RFQ template
- drawing-brief.md ← what the SVG drawings must show
- visual-bom-brief.md ← printable visual BOM brief
- wolfram-starter.wl ← parametric Wolfram source
- risks.md ← red-team risk register
- resources.md ← citations and references
- jig-decision.md ← three jig build/no-build calls
- photo-shotlist.md ← ten-shot photo plan



- capstone-deck.md / .pptx ← recruiter-facing slide deck
- print-packet.md / .html / .pdf ← shop-printable packet
- capstone-manifest.json ← deck-and-packet manifest
- xylophone-design-table.xlsx ← parametric source of truth
- images/ ← hero, build photos
- drawings/ ← per-bar SVGs + frame elevation/plan
- cad/ ← SolidWorks master (deferred)
- cnc/ ← cnc-plan.json + setup-sheet.md
- jigs/ ← jig sketches (deferred)
- site/ ← static build-log site

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# family-spec.csv

Project artifact.

| note | target_hz | bar_length | bar_width | bar_thickness | base_pos | tooltips | highlighted | predicted_f0              | mode             | spec        | resonator | mass | options       | notes |
|------|-----------|------------|-----------|---------------|----------|----------|-------------|---------------------------|------------------|-------------|-----------|------|---------------|-------|
| C5   | 523.25    | 16.683     | 1.500     | 0.875         | 3.737    | 12.946   | 523.25      | free-free Euler-Bernoulli | f15.22028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| C#5  | 554.37    | 16.208     | 1.500     | 0.875         | 3.631    | 12.578   | 554.37      | free-free Euler-Bernoulli | f14.86028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| D5   | 587.33    | 15.747     | 1.500     | 0.875         | 3.527    | 12.220   | 587.33      | free-free Euler-Bernoulli | f14.52028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| D#5  | 622.25    | 15.299     | 1.500     | 0.875         | 3.427    | 11.872   | 622.25      | free-free Euler-Bernoulli | f14.20028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| E5   | 659.26    | 14.863     | 1.500     | 0.875         | 3.329    | 11.534   | 659.26      | free-free Euler-Bernoulli | f13.89028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| F5   | 698.46    | 14.440     | 1.500     | 0.875         | 3.235    | 11.205   | 698.46      | free-free Euler-Bernoulli | f13.60028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| F#5  | 739.99    | 14.029     | 1.500     | 0.875         | 3.142    | 10.886   | 739.99      | free-free Euler-Bernoulli | f13.33028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| G5   | 783.99    | 13.630     | 1.500     | 0.875         | 3.053    | 10.577   | 783.99      | free-free Euler-Bernoulli | f13.08028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| G#5  | 830.61    | 13.242     | 1.500     | 0.875         | 2.966    | 10.275   | 830.61      | free-free Euler-Bernoulli | f12.83028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| A5   | 880.00    | 12.865     | 1.500     | 0.875         | 2.882    | 9.983    | 880.00      | free-free Euler-Bernoulli | f12.61028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| A#5  | 932.33    | 12.498     | 1.500     | 0.875         | 2.800    | 9.699    | 932.33      | free-free Euler-Bernoulli | f12.39028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| B5   | 987.77    | 12.143     | 1.500     | 0.875         | 2.720    | 9.423    | 987.77      | free-free Euler-Bernoulli | f12.19028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| C6   | 1046.50   | 11.797     | 1.500     | 0.875         | 2.643    | 9.154    | 1046.50     | free-free Euler-Bernoulli | f12.00028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| C#6  | 1108.73   | 11.461     | 1.500     | 0.875         | 2.567    | 8.894    | 1108.73     | free-free Euler-Bernoulli | f11.81028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| D6   | 1174.66   | 11.135     | 1.500     | 0.875         | 2.494    | 8.641    | 1174.66     | free-free Euler-Bernoulli | f11.64028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| D#6  | 1244.51   | 10.818     | 1.500     | 0.875         | 2.423    | 8.395    | 1244.51     | free-free Euler-Bernoulli | f11.48028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| E6   | 1318.51   | 10.510     | 1.500     | 0.875         | 2.354    | 8.156    | 1318.51     | free-free Euler-Bernoulli | f11.33028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| F6   | 1396.91   | 10.211     | 1.500     | 0.875         | 2.287    | 7.923    | 1396.91     | free-free Euler-Bernoulli | f11.19028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| F#6  | 1479.98   | 9.920      | 1.500     | 0.875         | 2.222    | 7.698    | 1479.98     | free-free Euler-Bernoulli | f11.05028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| G6   | 1567.98   | 9.638      | 1.500     | 0.875         | 2.159    | 7.479    | 1567.98     | free-free Euler-Bernoulli | f10.92028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| G#6  | 1661.22   | 9.363      | 1.500     | 0.875         | 2.097    | 7.266    | 1661.22     | free-free Euler-Bernoulli | f10.80028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| A6   | 1760.00   | 9.097      | 1.500     | 0.875         | 2.038    | 7.059    | 1760.00     | free-free Euler-Bernoulli | f10.69028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| A#6  | 1864.66   | 8.838      | 1.500     | 0.875         | 1.980    | 6.858    | 1864.66     | free-free Euler-Bernoulli | f10.58028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| B6   | 1975.53   | 8.586      | 1.500     | 0.875         | 1.923    | 6.663    | 1975.53     | free-free Euler-Bernoulli | f10.48028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |
| C7   | 2093.00   | 8.342      | 1.500     | 0.875         | 1.869    | 6.473    | 2093.00     | free-free Euler-Bernoulli | f10.38028(h/L^2) | stpt(E/rho) | solved    | defr | PadangeE#12.6 |       |

# jig-decision.md

*Project artifact.*

## Xylophone — Jig Decisions

Three jig decisions, each with build/no-build recommendation, the break-even calculation that drives it, and a one-paragraph build sketch when the answer is build.

### 1. Node-hole drilling jig — **\*\*BUILD\*\***

**\*\*Decision.\*\*** Build a fixed-fence drilling jig with two adjustable stops keyed to the 22.4% / 77.6% positions of the bar length.

**\*\*Rationale.\*\*** 25 bars × 2 holes = 50 holes. Each hole must land within  $\pm 1/32$ " of the node position or the bar's first mode picks up audible damping. Hand-marking 50 holes from a tape measure is unreliable at this tolerance. A jig with two flip-stops set against the bar end pays back the build time on bar #4 or #5.

**\*\*Sketch.\*\*** Plywood base 22"×6"×1/2", drill press fence along the back edge, two T-track flip-stops in a pair of 6" T-tracks routed flush. Each bar's two stop positions get pre-marked on a strip sticker per note — operator slides the bar against the front stop for the low-side hole, drills, flips the front stop down and the back stop up, drills the high-side hole. The stops are positioned from the bar's actual length, not a percent, so the operator never does math at the drill press.

### 2. Bar-end-shaving sled — **\*\*BUILD\*\***

**\*\*Decision.\*\*** Build a low-angle hand-plane shooting sled sized to the longest bar.

**\*\*Rationale.\*\*** Tuning happens by removing material from the bar ends — *\*not\** the underside (xylophone has no arch). End-shaving needs to be square to the bar's long axis and parallel to the opposite end, otherwise the bar develops a wedge that affects both

the fundamental tuning and the cord-hole alignment. A shooting sled makes both square and parallel automatic.

**Sketch.** Hardwood base 20"×6", 90° fence along one long edge, a stop block at the right end and a clamp wedge at the left. The bar slides against the fence; the plane runs along the base; the stop block prevents over-running the cut. A piece of 220 paper glued to the fence prevents skating. Build the fence dead-square to the base — measure with a machinist square before glue-up.

### 3. Frame-rail miter sled — **\*\*NO BUILD; use existing miter saw\*\***

**Decision.** Cut the rail-to-leg miters on the existing sliding compound miter saw. No jig.

**Rationale.** Eight 45° miters total (four leg ends × two each). A standard miter saw with a sharp 80T blade lands every miter at  $\pm 0.25^\circ$  easily, and the joints are reinforced by gusset plates anyway. Building a miter sled would cost a half-day and pay back nothing for an 8-cut job.

**Note for marimba sister repo.** A multi-octave marimba has 60+ miter cuts in the resonator-tube wells alone. The marimba repo should re-evaluate this decision and likely build a dedicated sled.

# photo-shotlist.md

*Project artifact.*

## Xylophone — Photo Shotlist

Ten shots that the build-log site, capstone deck, and visual BOM all draw from. Tied to the repo-level ``docs/photo-pipeline.md`` rules (neutral background, 5500K key light, sensor at instrument plane, no flash, JPEG + RAW). When a photo is captured, drop it under ``images/`` with the slug filename listed below — the deck and site templates will pick them up.

### 1. Hero — ``images/01-hero.jpg``

Full instrument on a clean dark-grey background, three-quarter angle from above-right, both mallets resting at the keyboard end of the frame. Lens normal-to-wide (35–50mm full frame). This is the README hero swap — until captured, ``images/xylophone-hero-placeholder.svg`` stands in.

### 2. Bar fan — ``images/02-bar-fan.jpg``

All 25 bars laid out flat on a white seamless, shortest-to-longest, 1/2" gap between each. Top-down. This image goes on the deck's "Family Plan" slide.

### 3. Node-hole detail — ``images/03-node-hole-detail.jpg``

Macro of the 1/8" node hole on the C5 bar with paracord threaded through. Lens 100mm macro. Light raking from one side to bring out the cord texture and the hole chamfer. Goes on the assembly-manual's node-suspension step.

### 4. Frame elevation — ``images/04-frame-elevation.jpg``

Side-on, eye-level, frame only (no bars), centered. Shot far enough back to be free of perspective distortion (~85mm full frame).

Used in `drawing-brief.md` as the dimension reference photo.

## 5. Mallet trio — `images/05-mallet-trio.jpg`

Hard rubber, lexan, and rosewood-head mallets laid parallel on a white seamless, heads aligned at the top. Top-down. Used in the `bom.csv`-driven visual BOM.

## 6. Suspension cord — `images/06-suspension-cord.jpg`

Three-quarter view of one full bar suspended on the cord at both node holes, with the cord ends terminating at the rail with the brass finish washers. Shows the geometry without the rest of the bars in frame. Used on the build-log site's "How the bars are suspended" section.

## 7. Tuner on bar — `images/07-tuner-on-bar.jpg`

The Korg OT-120 displayed mid-measurement on the A5 bar, mallet just having struck. Captures the validation workflow in one frame. Used on the validation slide of the deck.

## 8. Resonator stub — `images/08-resonator-stub.jpg` (deferred)

A single quarter-wave aluminum tube cut to the C5 resonator length (5.22"), held against the underside of the suspended C5 bar. Shot when v2 build adds resonators. Until then, this slot stays blank.

## 9. Moisture meter — `images/09-moisture-meter.jpg`

Pin meter inserted into the end-grain of a Padauk blank pre-glue-up, display showing 6.5–7.5% MC. Goes on the assembly-manual's "Verify moisture" step.

## 10. Finished assembly — `images/10-finished-assembly.jpg`

Three-quarter wide-angle of the finished xylophone in a play-ready position (mallets in player's hands, bars attached, frame on a

real floor). Eye-level standing. The "this thing works" portrait that closes the build-log site.

# resources.md

*Project artifact.*

## Xylophone — Resources

Citations, references, and the source-of-truth pointers that this packet relies on. Where the packet states a number, this file says \*where the number came from\*.

### Acoustic physics

- **Fletcher, Neville H., and Thomas D. Rossing.** *The Physics of Musical Instruments*, 2nd ed., Springer, 1998. Chapter 19 (Bars, Plates, and Membranes) — the canonical free-free Euler-Bernoulli derivation. Equations 19.7–19.13 yield the  $K_{\text{free-free}} / K_{\text{cantilever}} = (4.730/1.875)^2 \approx 6.36$  factor used throughout this packet.
- **Bork, Ingolf.** "Practical Tuning of Xylophone Bars and Resonators," *Applied Acoustics* 46 (1995): 103–127. Source for the 12th-overtone tuning convention and the bar-end-shaving correction shape. Bork's measured-vs-flat-bar second-mode ratio ( $\sim 2.756$ ) is the basis for the `first-mode-vs-second-mode-misalignment` risk.
- **Suits, B. H.** "Basic Physics of Xylophone and Marimba Bars." *American Journal of Physics* 69(7) (2001): 743–750. Clean derivation of the arch-undercut math the marimba sister repo will inherit; **explicitly cited** here as the "deferred-to-marimba" reference.
- `acoustic-models.md` (instrument-maker-v4 skill, `references/acoustic-models.md`). Free-Free Beams section is the primary citation — it both consolidates the formula in Tony's house units ( $h, L, E, \rho$ ) and carries the empirical-correction guard rules that govern when K-constants do and do not apply.

### Material constants

- **Wood Database** (Eric Meier, <https://www.wood-database.com/>).  $E$  and  $\rho$  values for African Padauk ( $E \approx 12.6 \text{ GPa}$ ,  $\rho \approx 745 \text{ kg/m}^3$ ),



Honduras Rosewood ( $E \approx 14.1$  GPa,  $\rho \approx 1000$  kg/m<sup>3</sup>), and Hard Maple ( $E \approx 12.6$  GPa,  $\rho \approx 705$  kg/m<sup>3</sup>). Cross-checked against the K-constant table in `acoustic-models.md`.

## Tuning workflow / measurement

- **`record\_measurement.py`** (instrument-maker-v4 skill, `scripts/record\_measurement.py`). Schema for `validation.csv` and the per-family corrections database. The `tuner\_make\_model`, `temp\_F`, `humidity\_RH\_percent`, and `cents\_target` columns in this packet's `validation.csv` map 1:1 to the script's CLI flags.
- **Korg OT-120** orchestral chromatic tuner.  $\pm 0.1$  cent accuracy in the C5–C7 range. Listed as the device of record in `bom.csv`.

## Empirical-correction guard rules

- **`acoustic-models.md`** "Empirical-correction guard rules" section. Cited in `design.md`'s Governing Model and in every prediction line of `family-spec.csv`. The summary in shop-friendly terms: K2 and end-corrections **do not apply** to bar idiophones; cantilever K-constants apply to free-free via the 6.36 $\times$  factor; everything is *first-order* until measured.

## Repository conventions

- **`tongue-drum/README.md`** (sister repo). README structure, cross-repo link block, magazine-baseline attribution pattern. This packet's `README.md` rewrites in the same shape.
- **`tongue-drum/cad/`** (sister repo). SolidWorks master-layout convention for parametric design tables. Cited in `cad/README.md` as the model the xylophone CAD will follow when built.
- **`golden-examples.md`** (instrument-maker-v4 skill, `references/golden-examples.md`). Mode A root-mode layout exemplar — every file in this packet sits at the repo root because that reference says so.

## Marimba/xylophone difference summary

The single best one-page summary of how xylophone and marimba differ:

**\*\*Suits 2001\*\*** (above). The salient points the packet operationalizes:

- Xylophone tunes the second mode to a 12th (3:1 ratio) above the fundamental; marimba tunes to two octaves (4:1).
- Xylophone bars are typically rosewood or padauk, smaller cross-section, often without arch undercut in low-end / educational instruments.
- Xylophone resonators (when present) are shorter quarter-wave tubes; marimba resonators are tuned at the bench by ear.
- Mallets are harder for xylophone; bars consequently must be harder wood and well-supported at the nodes.

Every "marimba-equivalent" callout in `design.md` traces back to this section.

# risks.md

*Project artifact.*

## Xylophone — Risks Register

Five categories, at least one entry each, every entry with an attached verification *\*test\** per the v4 red-team specialist contract.

### Acoustic

#### bar-cracking-from-hard-mallets

**\*\*Risk.\*\*** Hard rubber and lexan mallets, especially struck near the bar end rather than at the strike line, can crack a flat-bar Padauk bar within months. Hard mallets are appropriate for xylophone — but the failure mode is real and accelerates with low ambient humidity.

**\*\*Test.\*\*** Strike test: with the C5 bar (longest, most stress) clamped in the suspension cord, strike 100 times near the strike line and 50 times deliberately near the end with a hard rubber mallet. Inspect for hairline cracks under raking light. Repeat after 30 days at shop ambient. Pass = no visible cracks; fail = any crack.

#### first-mode-vs-second-mode-misalignment

**\*\*Risk.\*\*** First-order math predicts only  $f_1$ . The 12th-overtone xylophone tuning convention asks for the second flexural mode to sit at a 3:1 ratio, which a flat bar does **\*\*not\*\*** naturally do — the unpinched second-mode ratio of a free-free Euler-Bernoulli beam is closer to 2.756. Means players will hear the second mode "out of tune" with the fundamental.

**\*\*Test.\*\*** Spectrum-analyzer measurement (free phone app like SpectroidPro): pluck/strike the bar, capture the spectrum, identify the fundamental and the strongest second peak, compute the ratio. Pass = ratio between 2.7 and 3.0 (unundercut bars cluster around 2.756); fail = ratio outside that band. Document. The marimba sister repo will fix this with arch undercut; the xylophone v1 packet *\*accepts\** the natural ratio and notes it on the validation sheet.

## Structural

### moisture-retune-drift

**\*\*Risk.\*\*** Padauk at 6–8% MC at glue-up will move when the shop ambient changes. A 4% MC swing across a humid summer to a dry winter can shift fundamentals by 5–15 cents. Not a structural failure, but a tuning failure that compounds across all 25 bars.

**\*\*Test.\*\*** Record temp/RH at every measurement row in `validation.csv` (rows V2 through V8). Re-measure every bar at 30 and 60 days (rows V6, V7). Pass = drift within  $\pm 5$  cents on all bars at both checkpoints; fail = drift exceeds  $\pm 5$  cents — flag the bar in `record\_measurement.py`, update the per-family corrections database.

### cord-hole-tear-out

**\*\*Risk.\*\*** 1/8" node holes drilled near the bar's high-stress mode zone. Repeated player force on the cord at the hole can tear out the hole, especially if grain runs across it.

**\*\*Test.\*\*** Pull-test on the longest bar (C5): tie the suspension cord through both holes, suspend the bar, hang a 5-lb weight from the center of the bar, hold for 60 seconds. Inspect holes for cracking or elongation. Pass = no visible damage; fail = elongation, splitting, or cord chafing. If fail, drill a slightly larger hole and chamfer both ends.

## Ergonomic

### frame-tipping-on-uneven-floor

**\*\*Risk.\*\*** 4-leg trestle frame with a 21" rail and 30" leg height can tip if a player strikes near the rail end with a heavy two-handed attack on an uneven floor.

**\*\*Test.\*\*** Place frame on a 5° tilted board. Have a player strike the C5 bar (rail end) with full two-handed force using yarn-wrapped medium mallets. Frame should not tip. Pass = stable; fail = tips. Mitigation if fail: add 6" leg-foot crossbraces or weight the bottom of each leg.

## Supply

### **honduras-rosewood-cites-lead-time**

**\*\*Risk.\*\*** If a player overrides the Padauk default and chooses Honduras Rosewood, the species is **\*\*CITES Appendix II\*\*** as of 2017. Sourcing with proper documentation is slow (60–120 days) and expensive (~3× Padauk).

**\*\*Test.\*\*** If HR is selected, validate before order: supplier provides CITES export permit number; cross-check against the CITES species database. Pass = permit number valid and product description matches; fail = no permit or mismatch — reject the order, fall back to Padauk.

## Fit/Finish

### **tung-oil-finish-darkening-padauk**

**\*\*Risk.\*\*** Padauk is naturally bright orange-red but oxidizes to a deeper brown over months. Tung oil accelerates the oxidation. After 6–12 months bars look noticeably darker than at first build — visual mismatch if a single bar is replaced later.

**\*\*Test.\*\*** Cut a 2"×2" Padauk swatch at the same time as the bars. Apply same finish schedule. Store with the bars. After 30, 60, 180 days, photograph swatch under controlled light. Pass = expected gradual darkening, no spotting or blooming; fail = uneven oxidation patches — switch to a UV-blocking topcoat.